**Information Theory Mn908 and MICAS**

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**Exercise 2 of September 30, 2026**

(due on October 7, 2026, 8:30)

## Problem 1: Huffman Codes

Consider the following 8-ary codes:

- $\mathcal{C}_1 = \{10, 111, 101, 0110, 1010, 11100, 10010, 01001\}$
- $\mathcal{C}_2 = \{01, 000, 001, 100, 101, 1100, 1101, 1111\}$
- $\mathcal{C}_3 = \{10, 110, 111, 011, 010, 000, 0010, 0011\}$.

For each code $\mathcal{C}_i$, where $i = 1, 2, 3$, answer the following two questions:

(i) Is there an 8-ary source $X$ with probability distribution $(p_1, \ldots, p_8)$ such that $\mathcal{C}_i$ is a Huffman code for this source?

(ii) If such a source $X$ exists, what is such a probability distribution $(p_1, \ldots, p_8)$?

## Problem 2: Optimal Code Lengths that Require Almost One Bit above Entropy

The source coding theorem shows that the optimal code for a chance variable $X$ has an expected length less than $H(X) + 1$. Give an example of a chance variable for which the expected length of the optimal code is close to $H(X) + 1$, i.e., for any $\varepsilon > 0$, construct a distribution for which the optimal code has expected length larger than

$$H(X) + 1 - \varepsilon.$$

## Problem 3: Codes with Costs

Words like "Run!", "Help!", and "Fire!" are short, not because they are frequently used, but perhaps because time is precious in the situations in which these words are required. Suppose that $X = i$ with probability $p_i$, $i = 1, 2, \ldots, m$. Let $l_i$ be the number of binary symbols in the codeword associated with $X = i$, and let $c_i > 0$ denote the cost per letter of the codeword when $X = i$. Thus, the average cost $C$ of the description of $X$ is

$$C = \sum_{i=1}^{m} p_i c_i l_i.$$

1. Minimize $C$ over all $l_1, l_2, \ldots, l_m$ such that

$$\sum_{i=1}^{m} 2^{-l_i} \leq 1.$$

Ignore any implied integer constraints on $l_i$. Exhibit the minimizing $l_1^*, l_2^*, \ldots, l_m^*$ and the associated minimum value $C^*$.

2. How would you use the Huffman code procedure to minimize $C$ over all uniquely decodable codes? Let $C_{\text{Huffman}}$ denote this minimum.

3. Show that
$$C^* \leq C_{\text{Huffman}} < C^* + \sum_{i=1}^{m} p_i c_i.$$

## Problem 4: Entropy Rate

Let $\{X_i\}_{i \geq 1}$ be a stationary process. We define the processes $\{Y_i\}$, $\{Z_i\}$, and $\{V_i\}$ as follows:

$$Y_i = (X_i, X_{i+1}), \qquad Z_i = (X_{2i}, X_{2i+1}), \qquad V_i = X_{2i}.$$

1. Which of these processes are stationary?

2. Relate the entropy rates $H(\{Y_i\})$, $H(\{Z_i\})$, and $H(\{V_i\})$ to $H(\{X_i\})$.